

Canonical Build Sequence & Ecosystem Priorities

1. Executive Summary

This document codifies the phased implementation roadmap, bootstrapping dependencies, and closed-loop verification protocols for the NovÆxorpheus distributed edge AI ecosystem.

2. Phased Priority Roadmap

Priority 1: Architectural Grill & Scrutiny

- Interrogate documentation and establish non-destructive boundaries.
- Formalize repository boundaries (`aesop-xi`, `novus-aexenti`, `novaexopia`, `skills-and-capabilities`, `_dumbass_universal_memory`).

Priority 2: Ground Truth & Universal Tool Access

- Formalize Open Brain (OB1) protocol specifications over MCP.
- Establish Reasoning Bank ledger (`active_execution_paths.json`).
- Ensure all agents can access universal tooling: OpenWiki, Graphify AST, Obsidian vault, and NotebookLM-py.

Priority 3: Local Workspace Infrastructure & Clean Management

- Bootstrap repository file management protocols.
- Implement living LLM wiki schemas, Graphify AST parsers, and local Markdown vaults.
- Standardize the 3-file baseline per repository (`MAP.md`, `manifest.jsonl`, `README.md`).

Priority 4: In-Memory & Inference Deployment Layer

- Deploy and verify `mem0` in-session episodic state tracking.
- Operationalize OmniRoute gateway on `http://localhost:20128/v1` with RTK token compression and asynchronous memory extraction hooks.
- Establish hardware headroom thresholds for Snapdragon 8 Elite (Hexagon v79 NPU) and Jetson Orin Nano Super.

Priority 5: Native 3-APK Android Implementation & LMK Shielding

- Build and harden decoupled on-device daemons:
 - **Horizons UI (APK 1)**: Presentation shell and Chromium WebView.

- **Æsc (APK 2):** Terminal daemon running local ADB loopback on `127.0.0.1:5555` with UID 2000.
- **Æyre (APK 3):** Media and sensory daemon running Silero VAD, Moonshine STT, and Kokoro TTS.
- Verify process isolation to survive Android Low Memory Killer (LMK) cycles.

Priority 6: Full Convergence & Sovereign Execution

- Perfect WebSocket bridges and event bus handoffs.
- Align tri-model traces (Query, Executor, Frontier) with the incognito Red Auditor sandbox.
- Finalize the `#d.u.m.b.a.s.s.` unified memory bank.